

The Calculation of Ideal Gas

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Now we calculate the molar heat capacity of ideal gas in isochoric process, isobar process, isothermal process, adiabatic process and general polytropic process respectively.

1 Isochoric Process

By the definition of molar heat capacity:

$$c = \frac{dQ}{dT} \frac{1}{n}$$

we obtain that for isochoric process, the work from the environment $W = 0$, so by the first law of thermodynamics, we have:

$$c_V = n \frac{d\Delta U}{dT} \frac{1}{n}$$

So we obtain:

$$c_V = \frac{3}{2}R$$

where R is the constant of ideal gas.

2 Isobar Process

For any isobar process, the pressure is always a constant, denoted by p , so:

$$c_p = \frac{1}{n} \left(\frac{dU}{dT} + \frac{dW}{dT} \right)$$

As $dW = pdV$, we obtain:

$$c_p = \frac{1}{n} \left(\frac{dU}{dT} + p \frac{dV}{dT} \right)$$

or equivalently, we have:

$$c_p = c_V + \frac{p}{n} \frac{dV}{dT}$$

By the Law of Ideal Gas, we obtain:

$$c_p = c_V + R = \frac{5}{2}R$$

3 Isothermal Process

It is easy to obtain that for any isothermal process, we have the temperature T remains the same. At this time, $dT \rightarrow 0$, so the molar heat capacity is:

$$c_T \rightarrow \infty$$

4 Adiabatic Process

Obviously, $\Delta Q = cn\Delta T = 0$, however ΔT and n are all not zero, so we obtain only $c_Q = 0$. Now we consider a tiny period in the $p - V$ graph, we have:

$$dU = -dW = -pdV$$

And we know that:

$$dU = nc_V dT$$

Combining these two equations we can obtain:

$$nc_V dT + pdV = 0$$

By the Law of Ideal Gas, we obtain:

$$T = \frac{pV}{nR}$$

So we have:

$$\frac{c_V}{R} d(pV) = -pdV$$

By the formula of total differential, we have:

$$d(pV) = p dV + V dp$$

So after integration we obtain:

$$pV^{\frac{5}{3}} = C$$

5 General Polytropic Process

For the general polytropic process, they satisfy:

$$pV^k = C$$

we set the molar heat capacity of it is c_k , then by the first law of thermodynamics, we have:

$$dQ = nc_V dT + p dV$$

And we know:

$$c_k = \frac{1}{n} \frac{dQ}{dT} = c_V + p \frac{dV}{dT} \frac{1}{n}$$

By the Law of Ideal Gas, we obtain:

$$T = \frac{pV}{nR}$$
$$dT = \frac{1}{nR}d(pV)$$

So after integration, we obtain:

$$\frac{c_k - c_V}{R} - 1 \ln V = \frac{c_V - c_k}{R} \ln p$$

Then we have:

$$pV^{(1 - \frac{R}{c_k - c_V})} = C$$

So finally, we obtain:

$$c = c_V - \frac{R}{k - 1}$$